Zainab Raza

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Education

NED University Of Science And Technology — Karachi, Pakistan
Bachelor In Computer Information And System Engineering — *Oct 2022 – September 2026*

- Relevant Coursework: Database Management System, Machine Learning, Distributed Computing, Computer Vision, Artificial Intelligence

BAMM PECHS Govt. College For Women — Karachi, Pakistan
Intermediate (Pre-Engineering) — 86.7% — *2020 – 2022*

Technical Skills

Software Testing: SDLC, STLC, Functional Testing, Integration Testing, API Testing, Test Case Design, Requirement Analysis, Data Validation, Defect Tracking, Regression Testing
Testing Tools: Jira, pytest, Postman, Swagger/OpenAPI, Android Studio, Microsoft Excel(Test Cases, Pivot Tables)
Databases: MySQL, PostgreSQL.
Programming Languages: Python, JavaScript, HTML/HTML5, CSS, SQL, TypeScript
Frameworks & Development Tools: FastAPI, Flask, Django, ReactJS, Flutter, Git, GitHub, Visual Studio
Libraries & Technologies: PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, TensorFlow, Keras, Firebase, Supabase

Professional Experience

Xictek Systems -HisabDo (Finance Management App) — Aug 2026 – Sep 2026
AI/ML Intern — *Lahore, Pakistan*

- Contributed to AI solutions for software business applications.
- Developed and tested AI/API integration components using Python and pytest.
- Mapped and validated application use cases and AI service interactions.
- Worked on Generative AI/LLM-powered applications, prompt engineering, AI safety guardrails, and API integration.
- Implemented AI safety validation for emergency detection, diagnosis and prescription restrictions, prompt injection, and unsafe AI outputs.
- Used Jira for task and bug tracking and Git/GitHub version control system for development workflows.

National Centre For Big Data & Cloud Computing (NCBC), NED — Feb 2025 – Apr 2025
AI Intern — *Karachi, Pakistan*

- Collected, cleaned, validated, and analyzed datasets using Python to improve data quality and support machine learning experiments.
- Performed exploratory data analysis (EDA), data validation, and model evaluation to identify trends and improve model performance.
- Documented workflows, model architectures, experiments, and results.

Projects

AI-Driven Wearable Assistive Device for Navigation & Fall Detection (FYP Project)

- Developed an IoT wearable using ESP32, ultrasonic and IMU sensors, FastAPI, YOLO11, and LLM-based scene understanding for real-time assistive navigation. Integrated Generative AI and Vision-Language Models via Prompt Engineering to convert complex camera streams into natural audio descriptions.
- Implemented sensor-fusion algorithms for motion classification and adaptive power management, with priority-based TTS and haptic alerts delivered through Flutter.
- Built an automated emergency-response pipeline using Firebase and Supabase (PostgreSQL) for fall detection, real-time GPS tracking, and caregiver notifications.

Mood Based Music Recommendation System (Large Language Model System)

- Built a full-stack web application using React (Vite) and Flask that integrates Spotify Web API and Google Gemini API to generate personalized music recommendations. Implemented JWT authentication, PostgreSQL database integration, and RESTful API architecture.

Autonomous Game-Playing AI Agent (Flappy Bird)

- Developed an autonomous game-playing AI agent using Python, Pygame, and object-oriented programming. Implemented NEAT using neat-python to evolve neural network architectures, extracted spatial features for real-time decision-making and fitness evaluation, and tuned mutation rates, population size, and species survival parameters to optimize performance.

Air Quality Monitoring System (IoT Based System)

- Implemented an ESP32-based IoT monitoring system with DHT11 and MQ-135 sensor interfacing, real-time environmental data acquisition, OpenWeatherMap REST API integration, and ThingSpeak cloud telemetry for remote monitoring and analytics.

Martian Terrain Image Classification & Object Detection (Internship Project)

- Built a full-stack computer vision application integrating image classification (CNNs, InceptionNet, ResNet) and object detection (YOLO, Faster R-CNN) with real-time web deployment using React and JavaScript.

Certificates & Extracurricular Activities

- **Certificates:** Data Manipulation Using Python Libraries (Udemy), Google IT Automation with Python, and Google Data Analytics.
- **Extracurricular Activities:**
  - Contributed to AI Club projects focused on computer vision, multimodal AI, and real-time video understanding.
  - Worked on a CLIP-based surveillance system for semantic search and event understanding across video feeds.
  - Explored object detection, image-text embeddings, and model integration for practical AI applications.